UNDERSTANDING

# Alzheimer's *Disease*

What it is, how it progresses, and how we can **help**.



Presented by Ayse Chumber | Fall 2026

• WHAT IS ALZHEIMER'S DISEASE?

# A progressive brain disease that slowly damages *memory*, *thinking* and *behavior*.

OVER TIME, EVERYDAY TASKS BECOME HARDER

- Cooking
- Driving
- Managing money



## It is *not* a normal part of aging.

Forgetting a name now and then is normal. Alzheimer's is a disease.

IMPORTANT FACTS

# Three facts to *remember*

## #1

### most common cause of dementia

Dementia is the general term for losing memory and thinking skills. Alzheimer's causes most cases.

## 65+

### age when symptoms usually begin

Most people develop symptoms after age 65, but aging alone does not cause it.

## 0

### cures available today

Treatment can still help manage symptoms, and some new medicines can slow the disease.

• SIGNS AND SYMPTOMS

# The early signs are *easy to miss*

### Forgetting new information

e.g. forgets a conversation from this morning

### Repeating the same questions

e.g. asks “What day is it?” many times an hour

### Confusion about time or place

e.g. gets lost on a familiar street

### Trouble with familiar tasks

e.g. can’t follow a recipe they have made for years

### Difficulty finding words

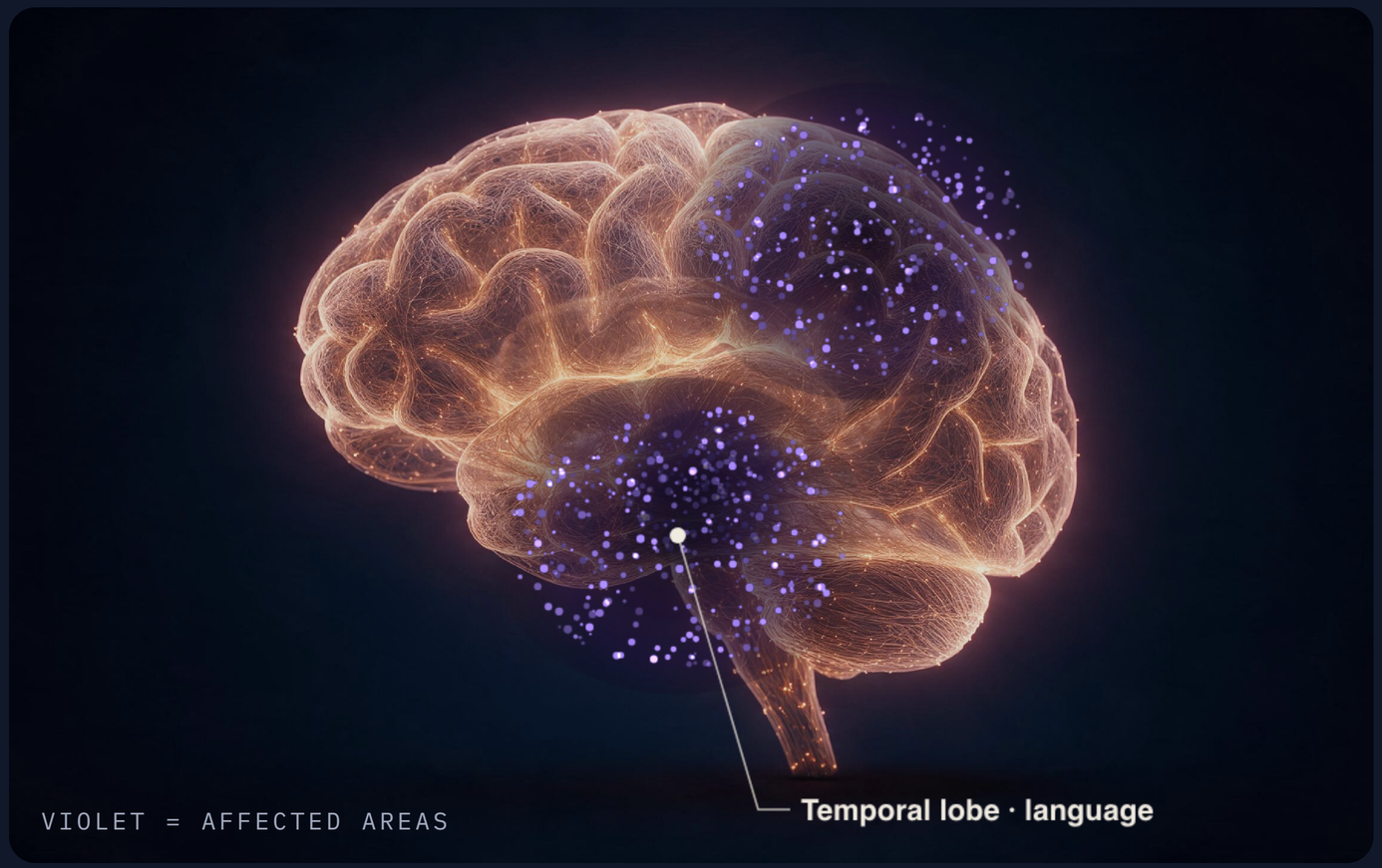
e.g. stops mid-sentence, can’t name everyday objects

### Changes in mood or personality

e.g. becomes anxious, suspicious or withdrawn

• STAGES

# Alzheimer's progresses in *three stages*




Early | **Middle** | Late

## Middle stage

Memory loss and confusion get worse. The person needs help with daily activities.

IN THE BRAIN

Damage spreads to areas for language, reasoning and the senses.

Independence — **Needs daily help**

• CAUSES AND RISK FACTORS

# The exact cause is unknown. Some factors *raise the risk*.

## Can’t be changed

### Age

The strongest risk factor, especially after 65.

### Genetics

Family history and the APOE-ε4 gene variant can increase risk.

## Can be influenced

HEALTHY HABITS HELP

### Heart & metabolic health

High blood pressure, diabetes, heart disease

### Smoking

Damages blood vessels, including the brain's

### Head injuries

Serious or repeated injuries

### Physical inactivity

Too little regular movement

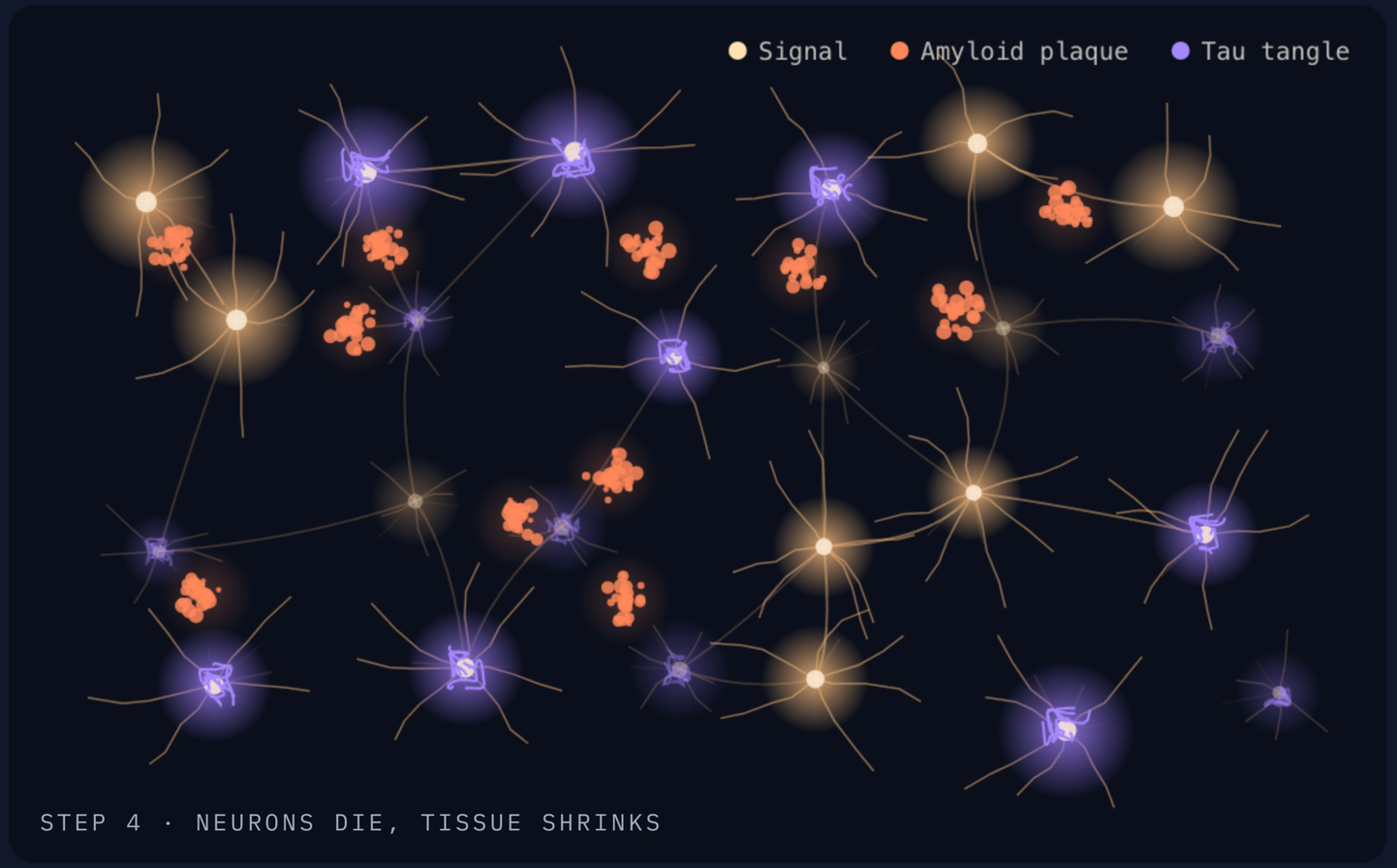
### Poor sleep

Long-term poor sleep quality

Source: Nystuen et al., 2024

• WHAT HAPPENS INSIDE THE BRAIN?

# Two proteins go wrong, and *neurons stop talking*




1 **Amyloid plaques** ●
Amyloid-beta protein clumps together between nerve cells.

2 **Tau tangles** ●
Tau protein twists into tangles inside nerve cells.

3 **Connections break**
Signals between neurons are disrupted.

4 **Cells die, brain shrinks**
Neurons die and parts of the brain shrink, causing memory loss.

Sources: Liu et al., 2024; Nystuen et al., 2024

• HOW ALZHEIMER'S IS DIAGNOSED

# No single test. Doctors *put the pieces together.*

LISTEN

## Medical & family history

Symptoms, medicines, family history

TEST

## Memory & thinking tests

Short tests of memory, language and attention

EXAMINE

## Physical & neurological exams

Plus blood tests to rule out other conditions

LOOK

## Brain scans

MRI, CT or PET show changes in the brain

CONFIRM

## Biomarker tests

Detect amyloid-beta and tau changes, when needed

**Why early?** An early diagnosis gives the person and family time to plan, and the newest medicines only work in the early stage.

Source: Liu et al., 2024

TREATMENT

# There is no cure yet, *but treatment helps*

## Medicine

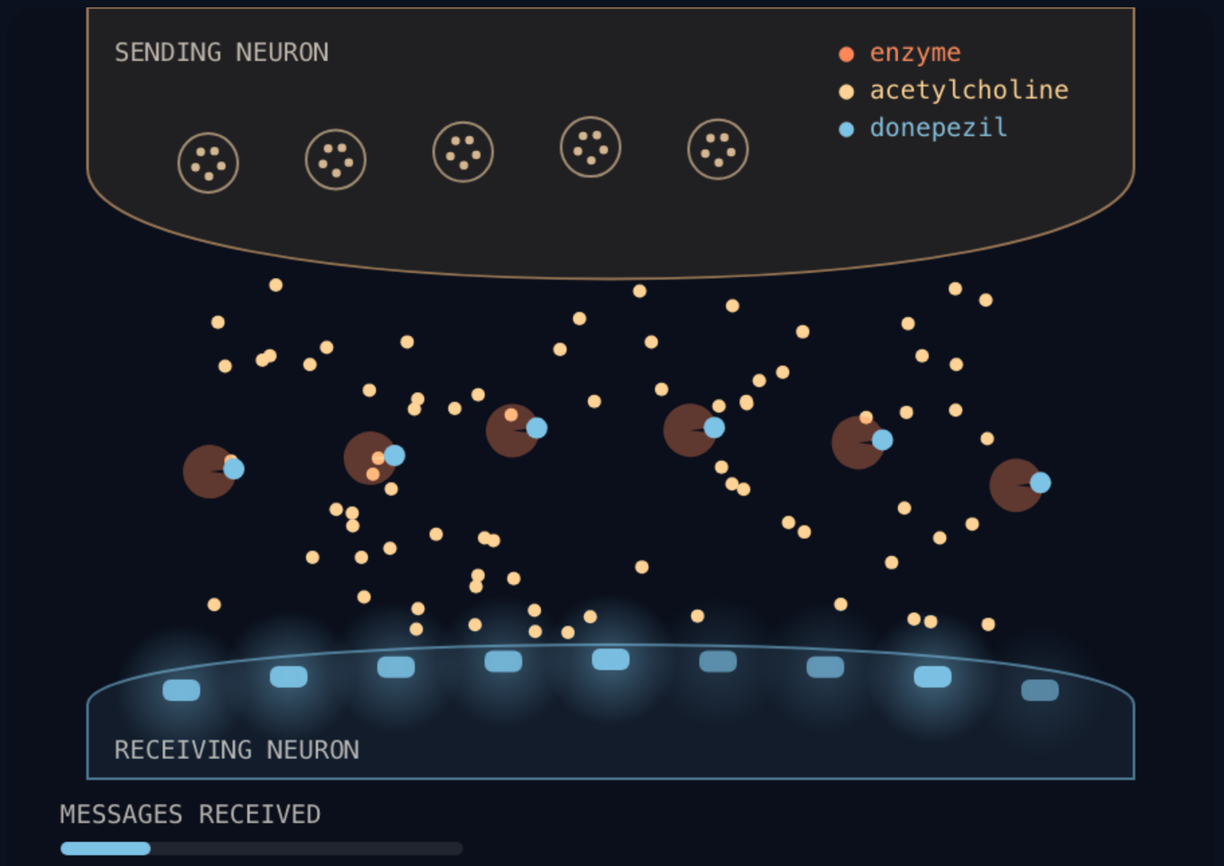
- **Symptom medicines**
  may temporarily improve memory and thinking.
- **New medicines**
  may slow the disease in some people in the early stage.

## Daily life

- **Daily routines**
  reduce confusion and stress.
- **Exercise and social activity**
  support overall health.
- **Family and caregivers**
  provide essential support.

• MEDICATIONS

# How the medicines *work*




Donepezil | Lecanemab · Donanemab

## Donepezil

BRAND NAME: ARICEPT

Stops an enzyme from breaking down acetylcholine, a chemical messenger for memory. More messages get through.

MAY TEMPORARILY HELP MEMORY AND THINKING

⚠ **Important:** none of these medicines cure Alzheimer's, they are not right for everyone, and all can have side effects. Lecanemab and donanemab need regular MRI scans to check for brain swelling or small bleeds.

Sources: van Dyck et al., 2023; Sims et al., 2023

SUPPORTING PATIENTS

# Small habits that make *a big difference*

### Speak calmly and simply

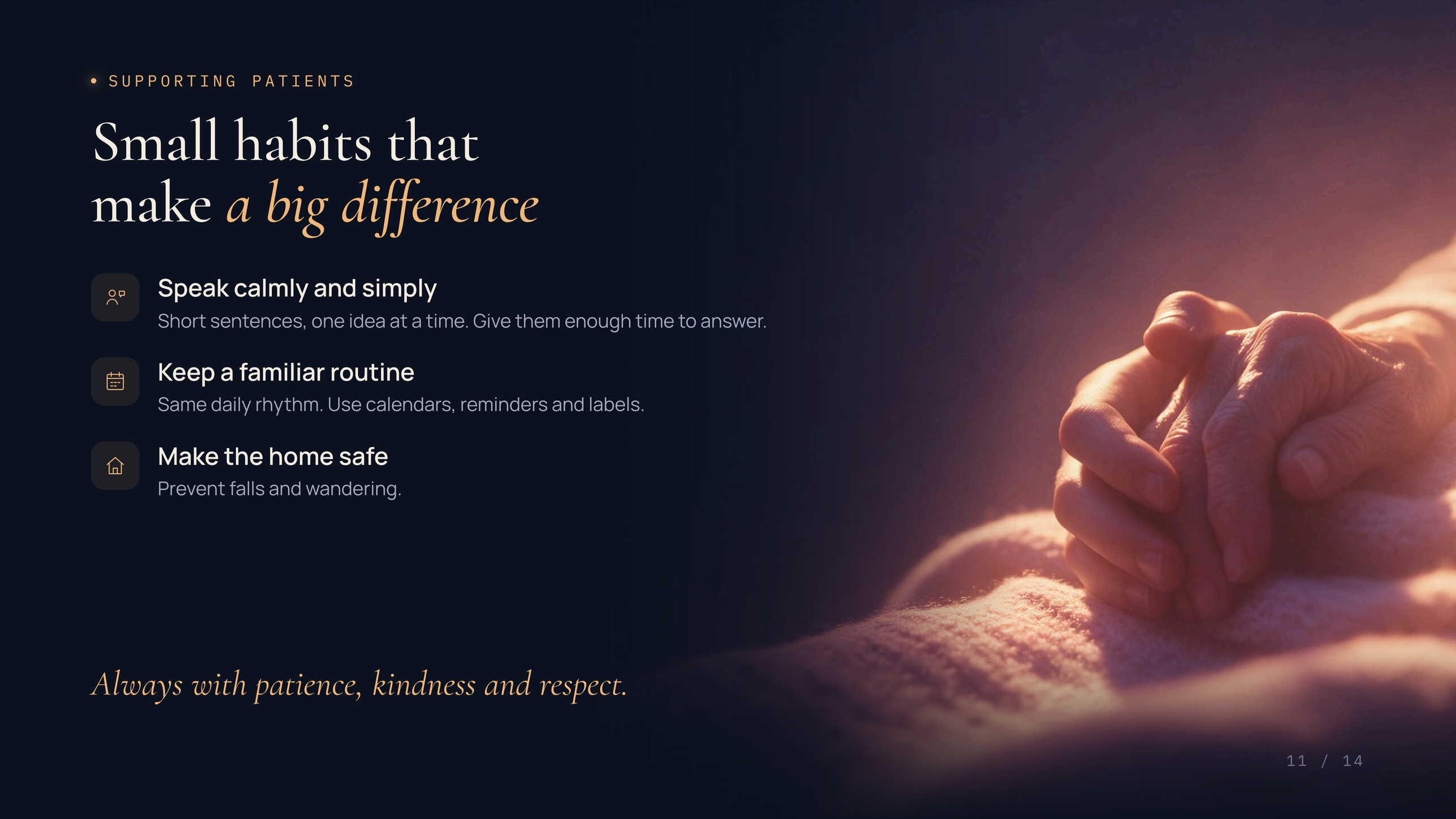
Short sentences, one idea at a time. Give them enough time to answer.

### Keep a familiar routine

Same daily rhythm. Use calendars, reminders and labels.

### Make the home safe

Prevent falls and wandering.



*Always with patience, kindness and respect.*

• CAREGIVERS

# Caregivers *need care too*



## Rest

Take regular breaks to avoid burnout.



## Share

Divide responsibilities among family members.



## Ask for help

Doctors, support groups and local services can help.



## Be patient

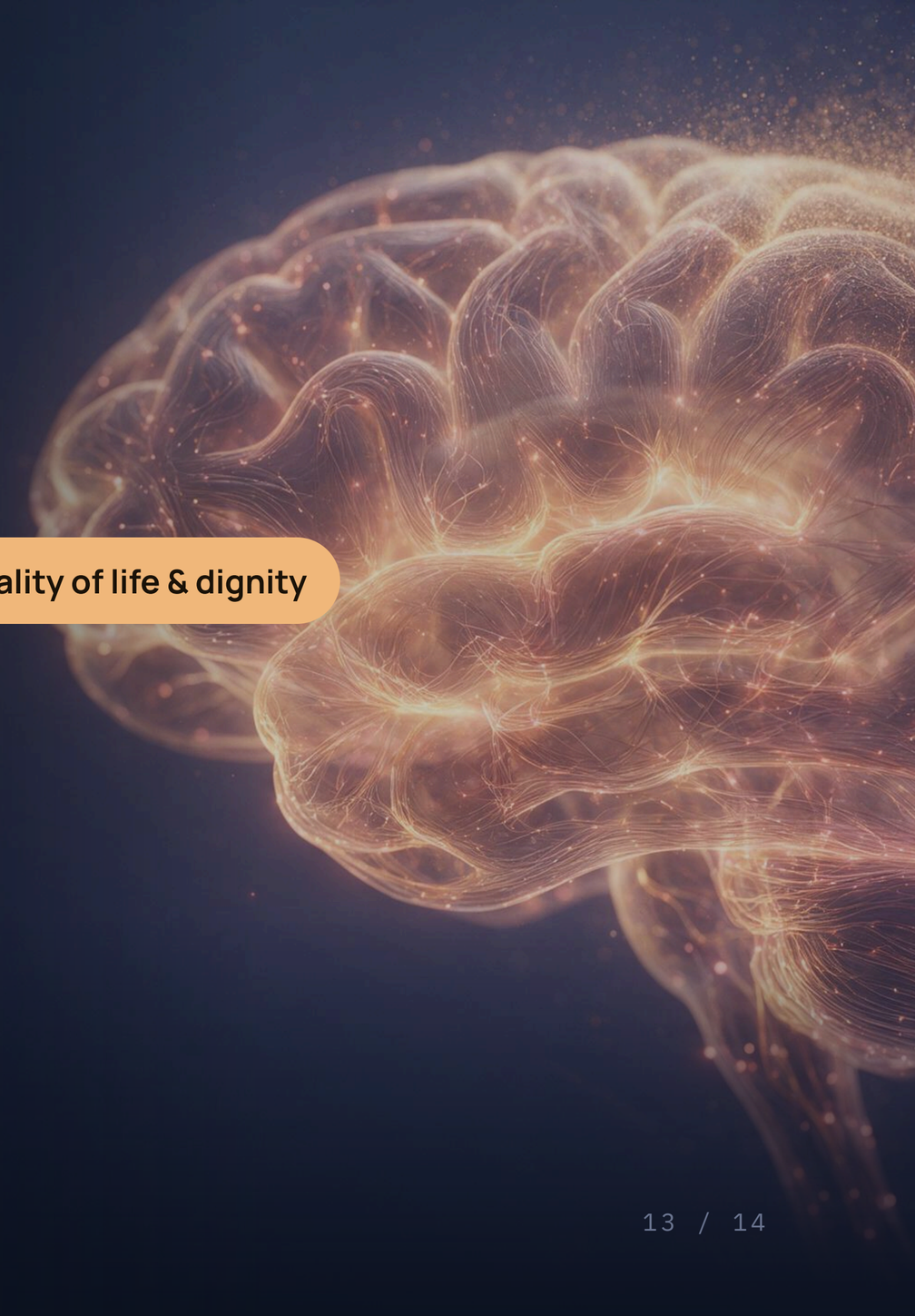
With the person you care for, and with yourself.

— A rested caregiver gives *better care*.

• CONCLUSION

# Alzheimer's changes the brain and everyday life, but *we can make a difference*.

Early diagnosis → The right treatment → Ongoing support = **Quality of life & dignity**



# *Thank you.*

Questions?

REFERENCES

# Based on *peer-reviewed* research


2024 Updates in Alzheimer's disease: From basic research to diagnosis and therapies

Liu, E., Zhang, Y., & Wang, J.-Z. *Translational Neurodegeneration*, 13, Article 45.

doi.org/10.1186/s40035-024-00432-x

2024 Alzheimer's disease: Models and molecular mechanisms informing disease and treatments

Nystuen, K. L., McNamee, S. M., Akula, M., Holton, K. M., DeAngelis, M. M., & Haider, N. B. *Bioengineering*, 11(1), Article 45.

doi.org/10.3390/bioengineering11010045

2023 Donanemab in early symptomatic Alzheimer disease: The TRAILBLAZER-ALZ 2 randomized clinical trial

Sims, J. R., Zimmer, J. A., Evans, C. D., et al. *JAMA*, 330(6), 512–527.

doi.org/10.1001/jama.2023.13239

2023 Lecanemab in early Alzheimer's disease

van Dyck, C. H., Swanson, C. J., Aisen, P., et al. *The New England Journal of Medicine*, 388(1), 9–21.

doi.org/10.1056/NEJMoa2212948